

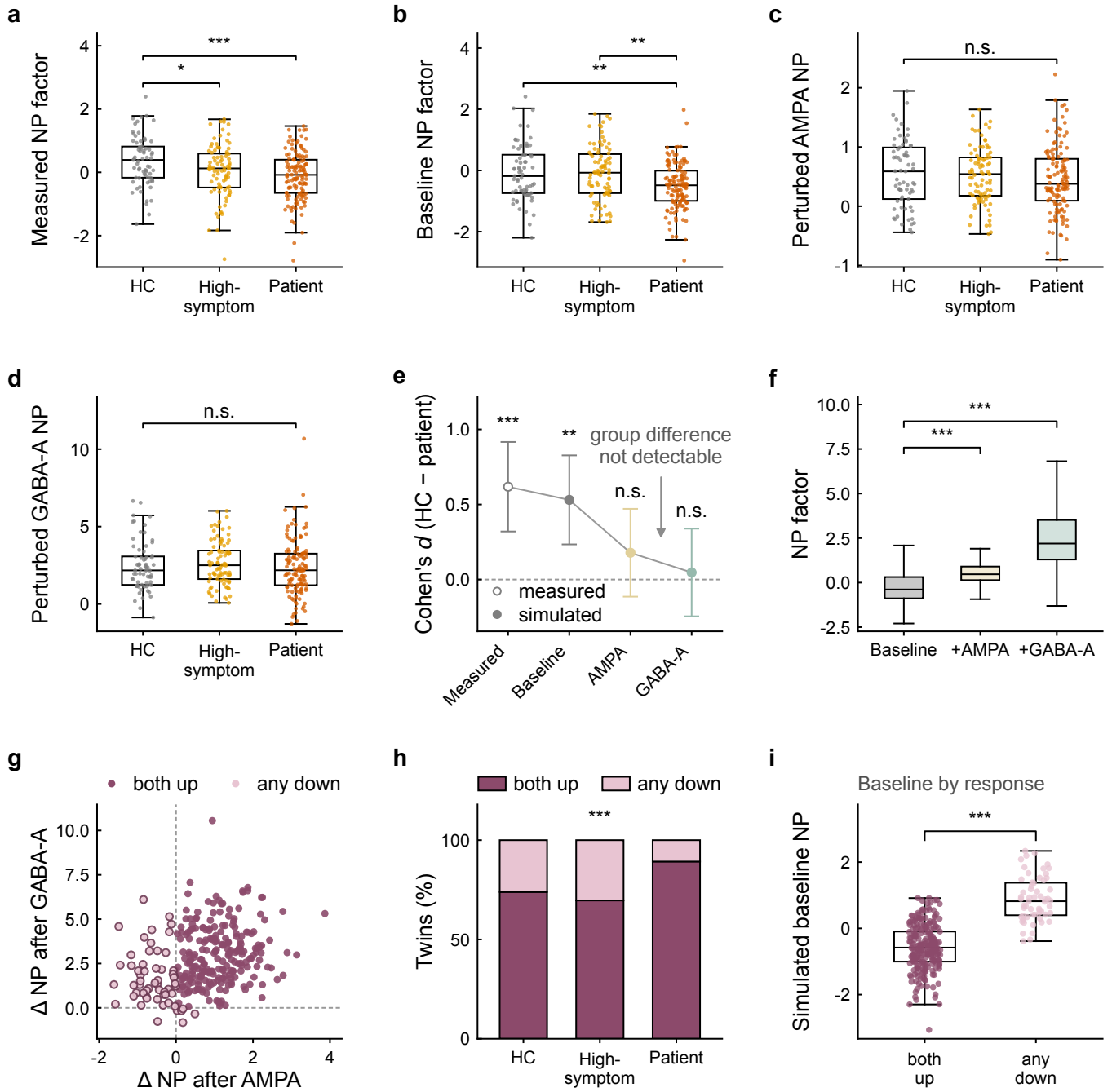


Fig. 4 | Virtual perturbation stratifies individuals. Cohort: $n = 288$ digital twins (HC 69, high-symptom 89, patients 130) after excluding two subjects with mean framewise displacement > 0.5 mm. GABA-A perturbation is always applied on top of the AMPA perturbation. **a–d**, NP factor by group in four conditions — measured (a), simulated baseline (b), after AMPA (c), after AMPA + GABA-A (d); each panel is scaled to its own data, so the within-panel contrast is comparable, not the absolute level. NP is residualised on sex, site and mean head motion, refit on these 288. Box plots show the median, 25th–75th percentiles and $1.5 \times$ IQR whiskers with all subjects overplotted; unit of observation, one subject. One-way ANOVA, $F(2, 285)$: a $F = 8.20$, $P = 0.00034$; b $F = 9.22$, $P = 0.00013$; c $F = 0.95$, $P = 0.39$; d $F = 0.85$, $P = 0.43$. Brackets are pairwise two-sided Welch t tests corrected within each panel by Benjamini–Hochberg FDR. Asterisks: * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$ in a–d (P in e–i); n.s., not significant. q for HC vs high-symptom / HC vs patient / high-symptom vs patient: a 0.024 / 0.0002 / 0.15; b 0.94 / 0.0016 / 0.001; c and d all $q \geq 0.44$, so both are marked n.s. **e**, HC – patient separation across the same four conditions (baseline = simulated baseline; AMPA / GABA-A = after that perturbation): Cohen's d with 95% CI, $n = 69$ HC versus 130 patients (the high-symptom group is excluded from this contrast by design); two-sided independent t tests, $df = 197$, uncorrected. The separation present in the measured data and reproduced by the twins is no longer detectable after either perturbation: $d = 0.62$ (95% CI 0.32 to 0.92), $t = 4.11$, $P < 0.0001$ measured; 0.53 (0.23 to 0.83), $t = 3.36$, $P = 0.001$ at baseline; 0.18 (–0.11 to 0.47), $t = 1.19$, $P = 0.23$ after AMPA; 0.05 (–0.24 to 0.34), $t = 0.33$, $P = 0.74$ after GABA-A. **f**, within-subject effect of each perturbation on the simulated network: box plots (median, 25th–75th percentiles, $1.5 \times$ IQR whiskers) of the 12-edge NP factor at baseline and after each perturbation; the axis is cropped to the whiskers and 7 of 864 values beyond them are not shown. Brackets are two-sided paired t tests against baseline (pairing on subject), $n = 288$, $df = 287$, uncorrected, with Cohen's d_z and the share of twins that increase: AMPA $+0.78 \pm 0.93$, $t = 14.14$, $d_z = 0.83$, 234/288 up; GABA-A $+2.74 \pm 1.71$, $t = 27.11$, $d_z = 1.60$, 282/288 up; both $P < 0.0001$. **g**, each twin's change in NP factor under AMPA against the change under GABA-A; 229 of 288 (79.5%) increase under both, 59 have at least one decrease (234/288 increase under AMPA, 282/288 under GABA-A). Descriptive classification; no test is applied to this split, because the grouping is defined by the sign of the plotted quantity. **h**, the response pattern by clinical group ($\chi^2(2) = 14.2$, $P = 0.00084$); both up / total: HC 51/69, high-symptom 62/89, patients 116/130, i.e. 89% of patients versus 74% of HC. **i**, simulated baseline NP in the two response groups (both up $n = 229$, any down $n = 59$): -0.57 versus 0.88 , two-sided Welch $t = -14.7$, $P < 0.0001$. The grouping is defined on the perturbation response and the plotted quantity is the unperturbed baseline, so this contrast is not circular.